

GRACE V. RINGLEIN

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EDUCATION

- PhD Candidate in Biostatistics**, Johns Hopkins Bloomberg School of Public Health 2022–2027 (Expected)
- Master of Science in Physics & Astronomy**, University of Pennsylvania 2017–2020
- Bachelor of Arts in Physics**, University of Pennsylvania 2016–2020
Minor in Fine Arts, Concentration in Computer Techniques
- Summa Cum Laude; Phi Beta Kappa; with Honors Distinction in Physics
 - Graduate of the Roy and Diana Vagelos Program for the Molecular Life Sciences

RESEARCH EXPERIENCE

JOHNS HOPKINS BLOOMBERG SCHOOL OF PUBLIC HEALTH

- Doctoral Research**, Department of Biostatistics Dec. 2022–
Advisors: Elizabeth Stuart, PhD; Trang Nguyen, PhD Present
- Dissertation topic: Causal inference under measurement error and unobserved confounding: methods and applications in electronic health records research.
- Collaborative Research**, Department of Health Policy and Management Dec. 2022–
Key Collaborator: Catherine Ettman, PhD Present
- Analysis and ongoing statistical support for investigations of the effect of job loss on mental health (and the potential moderating role of financial strain) using national survey data.
- Collaborative Research**, Precision Medicine Center of Excellence in Mood Disorders Jun. 2023–
Key Collaborators: Peter Zandi, PhD; Catherine Ettman, PhD Present
- Cohort construction and power analyses supporting Dr. Ettman's K01 grant *Economic Precarity, Costs, and Continuity of Mental Health Care* (now funded).
 - Analysis and ongoing statistical support for research projects related to access and continuity of mental health care using electronic health records.
 - Statistical support for projects related to comparative effectiveness of antidepressants.
- Collaborative Research**, Department of Epidemiology Aug. 2025–
Key Collaborator: John Jackson, ScD Present
- Extraction and synthesis for a literature review of health disparity models.

NATIONAL INSTITUTE OF MENTAL HEALTH

- Post-baccalaureate Fellow (IRTA)**, Section on Development and Affective Neuroscience 2020–2022
Principal Investigator: Daniel Pine, MD
- Co-authored publications related to pediatric mental health using neuroimaging data.
 - Conducted early data analysis and analytic planning for projects applying large language models to classify behaviors in therapy session transcripts.

UNIVERSITY OF PENNSYLVANIA

Undergraduate Research, Multiple laboratories

2017–2020

- Irina Marinov, PhD (Earth & Environmental Science): Investigated convection dynamics in Southern Ocean climate models.
- Emily Falk, PhD (Annenberg School of Communication): Created novel stimuli for MRI studies of addiction.
- Masao Sako, PhD (Physics & Astronomy): Constructed a pipeline for detecting microlensing events in Dark Energy Survey data.

PUBLICATIONS

PREPRINTS & UNDER REVIEW

1. **Ringlein, G. V.**, Nguyen, T. Q., Zandi, P. P., Stuart, E. A., & Parikh, H. Demystifying proximal causal inference. Under revision with *Statistical Science*. doi:10.48550/arxiv.2512.24413.
2. Ettman, C. K., Jeon, K., Dohlmán, P., **Ringlein G. V.**, Straub, J., et al. Psychiatric services following suicide-related hospital visits: An analysis of electronic health records among patients with depression. Under revision with *Psychiatric Research and Clinical Practice*.

SELECTED PUBLICATIONS

1. Nguyen, T. Q., Nabi, R., Yang, F., **Ringlein, G. V.**, & Stuart, E. A. Self-separated and self-connected models for mediator and outcome missingness in mediation analysis. *Statistical Science*. In press. <https://arxiv.org/abs/2411.07221>
2. Mady L. J. , **Ringlein G. V.**, Satpathy-Horton R., Stuart E. A., Galea S., Ettman C. K. Pre-existing financial circumstance, job loss, and probable depression among U.S. adults: a prospective population-based cohort study. *Lancet Reg Health Am*. In press.
3. **Ringlein, G. V.**, Ettman, C. K., & Stuart, E. A. (2024). Income or job loss and psychological distress during the COVID-19 pandemic. *JAMA Network Open*, 7(7), e2424601. doi:10.1001/jamanetworkopen.2024.24601
4. Ettman, C. K., **Ringlein, G. V.**, Dohlmán, P., Straub, J., Brantner, C. L., Chin, E. T., et al. (2025). Trends in mental health care and telehealth use across area deprivation: An analysis of electronic health records from 2016–2024. *PNAS Nexus*, pgaf016. doi:10.1093/pnasnexus/pgaf016
5. Hong, H., Zhao, C., Jeon, K., Goicoechea, E. B., Brantner, C. L., **Ringlein, G.**, et al. Comparative effectiveness of antidepressants for depression using EHRs from two health systems. *BMC Psychiatry* (2026). doi:10.1186/s12888-026-08159-x
6. Ettman, C. K., Brantner, C. L., Goicoechea, E. B., Dohlmán, P., **Ringlein, G. V.**, Straub, J., et al. (2025). Gaps in psychiatric care before and after the COVID-19 pandemic among patients with depression using electronic health records. *Psychiatry Research*, 344, 116354.
7. Brantner, C. L., Yu, W., Zhao, C., Jeon, K., **Ringlein, G. V.**, Wang, Q., et al. (2025). The challenges of integrating diverse data sources: A case study in major depression. *Health Services and Outcomes Research Methodology*. doi:10.1007/s10742-025-00363-8

ADDITIONAL PUBLICATIONS

8. Richier, C., Zugman, A., Harrewijn, A., Cardinale, E., Khosravi, P., et al. Brain Age Prediction in Generalized Anxiety Disorder using a Convolutional Neural Network. *Translational Psychiatry* 16, 368 (2026). doi:10.21203/10.1038/s41398-026-04078-3
9. Lossio-Ventura, J. A., Frank, S., **Ringlein, G.**, Bonson, K., Olszko, A., Knobel, A., et al. (2025). Automated classification of exposure and encourage events in speech data from pediatric OCD treatment. *JAMIA Open*, 8(6), ooaf127. doi:10.1093/jamiaopen/ooaf127

10. Zugman, A., **Ringlein, G. V.**, Finn, E. S., Lewis, K. M., Berman, E., Silverman, W. K., et al. (2025). Brain functional connectivity and anatomical features as predictors of cognitive behavioral therapy outcome for anxiety in youths. *Psychological Medicine*, 55, e91. doi:10.1017/S0033291724003131
11. Kirk, P. A., Qamar, P., Zugman, A., Abend, R., Frank, S., **Ringlein, G. V.**, et al. (2025). The relations among anxiety, movie-watching, and in-scanner motion. *Human Brain Mapping*, 46(4), e70163. doi:10.1002/hbm.70163
12. Medel, V., Alliende, L. M., Bethlehem, R., Seidlitz, J., **Ringlein, G.**, Arango, C., et al. (2025). Human development, inequality, and their associations with brain structure across 29 countries. *European Psychiatry*, 68(1), e100. doi:10.1192/j.eurpsy.2025.10060
13. Bruin, W. B., Zhutovsky, P., van Wingen G., Bas-Hoogendam, J. M., Groenewold, N. A., et al. Brain-Based Classification of Youth with Anxiety Disorders: an ENIGMA-ANXIETY Transdiagnostic Examination using Machine Learning. *Nature Mental Health* 2, 104–118 (2024). doi:10.1038/s44220-023-00173-2
14. Zugman, A., Alliende, L. M., Medel, V., Bethlehem, R. A. I., Seidlitz, J., **Ringlein, G.**, et al. (2023). Country-level gender inequality is associated with structural differences in the brains of women and men. *Proceedings of the National Academy of Sciences*, 120(20), e2218782120. doi:10.1073/pnas.2218782120
15. Ettman, C. K., Fan, A. Y., Philips, A. P., Adam, G. P., **Ringlein, G.**, Clark, M. A., et al. (2023). Financial strain and depression in the U.S.: A scoping review. *Translational Psychiatry*, 13(1), 168. doi:10.1038/s41398-023-02460-z
16. Harrewijn, A., Cardinale, E. M., Groenewold, N. A., Bas-Hoogendam, J. M., Aghajani, M., Hilbert, K., et al. (2021). Cortical and subcortical brain structure in generalized anxiety disorder: Findings from 28 research sites in the ENIGMA-Anxiety Working Group. *Translational Psychiatry*, 11(1), 502. doi:10.1038/s41398-021-01622-1
17. Gnanadesikan, A., Speller, C. M., **Ringlein, G.**, San Soucie, J., Thomas, J., & Pradal, M.-A. (2020). Feedbacks driving interdecadal variability in southern ocean convection in climate models: A coupled oscillator mechanism. *Journal of Physical Oceanography*, 50(8), 2227–2249. doi:10.1175/JPO-D-20-0037.1

HONORS AND AWARDS

Society for Causal Inference Travel Award , American Causal Inference Conference	2026
Lodging Award , Statistics in the Age of AI Conference, George Washington University	2024
Best Poster Award , Society for Epidemiologic Research Annual Meeting	2024
• “Trends in the Use of Telehealth by Area Deprivation in a Large U.S. Academic Hospital” (presented by Catherine Ettman).	
Roy and Diana Vagelos Science Challenge Award , University of Pennsylvania	2019
NASA Pennsylvania Space Grant Consortium Scholarship ,	2018
Francis A. Velay Fellowship , University of Pennsylvania	2017

TEACHING EXPERIENCE

JOHNS HOPKINS BLOOMBERG SCHOOL OF PUBLIC HEALTH

Teaching Academy Certificate Program , Johns Hopkins University Center for Teaching Excellence and Innovation	2025–Present
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Teaching Assistant , Department of Biostatistics	2023–Present
• <i>Causal Inference in Medicine and Public Health I</i> (3rd, 4th terms: 2024, 2025; 3rd 2026) – Guest lecture: Regression Discontinuity Designs (March 4, 2025; March 3, 2026) – Developing and teaching weekly tutorials • <i>Multilevel and Longitudinal Models: Data Analysis Workshop</i> (2nd term, 2025) • <i>Analysis of Multilevel and Longitudinal Models</i> (1st term, 2025) • <i>Statistical Methods in Public Health I–II</i> (1st and 2nd terms, 2024) • <i>Statistical Methods in Public Health IV</i> (4th term, 2026) • <i>Advanced Methods in Biostatistics I–II</i> (1st and 2nd terms, 2023) • Individual tutoring for a Biostatistics PhD student preparing for comprehensive exams (2025–26)	

UNIVERSITY OF PENNSYLVANIA

Teaching Assistant , Department of Physics and Astronomy	2018–2020
• <i>Laboratory Electronics</i> (Spring 2020) • <i>Physics for Architects</i> (Fall 2019) • Laboratory Instructor, <i>General Physics: Electricity, Magnetism, Optics, Modern Physics</i> (Fall 2019; Spring 2018)	

COMMUNITY, LEADERSHIP, & SERVICE

Biostatistics Student Organization , Johns Hopkins Bloomberg School of Public Health	2023–2025
• Mentorship Committee Co-lead (2025–26) • Curriculum Committee Representative for the PhD Program (2023–24, 2024–25) • Mental Health Graduate Network Representative for Biostatistics (2023–24, 2024–25)	
Alumni Mentor , Pennsylvania Governor’s School for the Sciences	2022–2023
Planetarium Presenter , Maryland Science Center	Summer 2022
Community Arts Instructor , Baltimore ClayWorks	Summer 2022
Alumni Interviewer , University of Pennsylvania	2020–2022
Research Assistant Onboarding Co-lead , Section on Development and Affective Neuroscience, NIMH	Summer 2021
Peer Advisor , University of Pennsylvania	2019–2020

TECHNICAL SKILLS

- **Programming:** R, Python, Bash, MATLAB, SQL, Java
- **Graphics:** Adobe Photoshop, Adobe Illustrator